Reference angles and trigonometric functions

1 Consider the trigonometric ratio $\sin 135^\circ$.

- **a** State which quadrant 135° is located.
- **b** Find the measure of the positive acute angle that 135° is related to.
- **c** Is $\sin 135^\circ$ positive or negative?
- **d** Rewrite $\sin 135^\circ$ in terms of its reference angle.

2 Consider the trigonometric ratio $\cos 240^\circ$.

- **a** State which quadrant 240° is located.
- **b** Find the measure of the positive acute angle that 240° is related to.
- **c** Is $\cos 240^\circ$ positive or negative?
- **d** Rewrite $\cos 240^\circ$ in terms of its reference angle.

3 Consider the trigonometric ratio $\tan 210^\circ$.

- **a** State which quadrant 210° is located.
- **b** Find the measure of the positive acute angle that 210° is related to.
- **c** Is $\tan 210^\circ$ positive or negative?
- **d** Rewrite $\tan 210^\circ$ in terms of its reference angle.

4 Consider the trigonometric ratio $\sin \dfrac{7\pi}{6}$.

- **a** State which quadrant $\dfrac{7\pi}{6}$ is located.
- **b** Find the measure of the positive acute angle that $\dfrac{7\pi}{6}$ is related to.
- **c** Is $\sin \dfrac{7\pi}{6}$ positive or negative?
- **d** Rewrite $\sin \dfrac{7\pi}{6}$ in terms of its reference angle.

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5 Consider the trigonometric ratio $\cos \frac{11\pi}{6}$.

a State which quadrant $\frac{11\pi}{6}$ is located.

b Find the measure of the positive acute angle that $\frac{11\pi}{6}$ is related to.

c Is $\cos \frac{11\pi}{6}$ positive or negative?

d Rewrite $\cos \frac{11\pi}{6}$ in terms of its reference angle.

6 Consider the trigonometric ratio $\tan \frac{5\pi}{4}$.

a State which quadrant $\frac{5\pi}{4}$ is located.

b Find the measure of the positive acute angle that $\frac{5\pi}{4}$ is related to.

c Is $\tan \frac{5\pi}{4}$ positive or negative?

d Rewrite $\tan \frac{5\pi}{4}$ in terms of its reference angle.

7 Consider the trigonometric ratio $\tan \frac{11\pi}{6}$.

a State which quadrant $\frac{11\pi}{6}$ is located.

b Find the measure of the positive acute angle that $\frac{11\pi}{6}$ is related to.

c Is $\tan \frac{11\pi}{6}$ positive or negative?

d Rewrite $\tan \frac{11\pi}{6}$ in terms of its reference angle.

8 Consider the trigonometric ratio $\cos \frac{7\pi}{4}$.

a State which quadrant $\frac{7\pi}{4}$ is located.

b Find the measure of the positive acute angle that $\frac{7\pi}{4}$ is related to.

c Is $\cos \frac{7\pi}{4}$ positive or negative?

d Rewrite $\cos \frac{7\pi}{4}$ in terms of its reference angle.

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9 Consider the trigonometric ratio $\sin \frac{4\pi}{3}$.

a State which quadrant $\frac{4\pi}{3}$ is located.

b Find the measure of the positive acute angle that $\frac{4\pi}{3}$ is related to.

c Is $\sin \frac{4\pi}{3}$ positive or negative?

d Which of the following is $\sin \frac{4\pi}{3}$ equivalent to: $-\sin \frac{\pi}{3}$ or $\sin \frac{\pi}{3}$?

10 Consider the trigonometric ratio $\cos \frac{4\pi}{3}$.

a State which quadrant $\frac{4\pi}{3}$ is located.

b Find the measure of the positive acute angle that $\frac{4\pi}{3}$ is related to.

c Is $\cos \frac{4\pi}{3}$ positive or negative?

d Which of the following is $\cos \frac{4\pi}{3}$ equivalent to: $-\cos \frac{\pi}{3}$ or $\cos \frac{\pi}{3}$?

11 Consider the trigonometric ratio $\tan \frac{7\pi}{4}$.

a State which quadrant $\frac{7\pi}{4}$ is located.

b Find the measure of the positive acute angle that $\frac{7\pi}{4}$ is related to.

c Is $\tan \frac{7\pi}{4}$ positive or negative?

d Which of the following is $\tan \frac{7\pi}{4}$ equivalent to: $\tan \frac{\pi}{4}$ or $-\tan \frac{\pi}{4}$?

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Exact values of trigonometric functions

12 Consider the angle 810°.

- **a** State the coordinates of the point on the unit circle that corresponds to 810°.
- **b** Find the value of $\cos 810^\circ$.
- **c** Find the value of $\sin 810^\circ$.
- **d** Determine whether the following are undefined:

  i $\cot 810^\circ$ **ii** $\csc 810^\circ$ **iii** $\sec 810^\circ$ **iv** $\tan 810^\circ$

13 Consider the angle (-360°).

- **a** State the coordinates of the point on the unit circle that corresponds to (-360°).
- **b** Find the value of $\cos(-360^\circ)$.
- **c** Find the value of $\sin(-360^\circ)$.
- **d** Determine whether the following are undefined:

  i $\cot(-360^\circ)$ **ii** $\csc(-360^\circ)$ **iii** $\sec(-360^\circ)$ **iv** $\tan(-360^\circ)$

14 Consider the angle 270°.

- **a** State the coordinates of the point on the unit circle that corresponds to 270°.
- **b** Find the value of $\cos 270^\circ$.
- **c** Find the value of $\sin 270^\circ$.
- **d** Determine whether the following are undefined:

  i $\tan 270^\circ$ **ii** $\cot 270^\circ$ **iii** $\csc 270^\circ$ **iv** $\sec 270^\circ$

15 Consider the angle (-180°).

- **a** State the coordinates of the point on the unit circle that corresponds to (-180°).
- **b** Find the value of $\cos(-180^\circ)$.
- **c** Find the value of $\sin(-180^\circ)$.
- **d** Determine whether the following are undefined:

  i $\sec(-180^\circ)$ **ii** $\cot(-180^\circ)$ **iii** $\csc(-180^\circ)$ **iv** $\tan(-180^\circ)$

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16 Consider the angle 720°.

a State the coordinates of the point on the unit circle that corresponds to 720°.

b Find the value of $\cos 720^\circ$.

c Find the value of $\sin 720^\circ$.

d Determine whether the following are undefined:

i $\sec 720^\circ$ **ii** $\cot 720^\circ$ **iii** $\tan 720^\circ$ **iv** $\csc 720^\circ$

17 Consider the angle 1260°.

a State the coordinates of the point on the unit circle that corresponds to 1260°.

b Find the value of $\cos 1260^\circ$.

c Find the value of $\sin 1260^\circ$.

d Determine whether the following are undefined:

i $\csc 1260^\circ$ **ii** $\cot 1260^\circ$ **iii** $\tan 1260^\circ$ **iv** $\sec 1260^\circ$

18 Find the exact value of the following trigonometric ratios:

a $\sin 330^\circ$ **b** $\cos 330^\circ$ **c** $\tan 330^\circ$ **d** $\csc 330^\circ$

e $\sec 330^\circ$ **f** $\cot 330^\circ$ **g** $\sin 510^\circ$ **h** $\cos 510^\circ$

i $\tan 510^\circ$ **j** $\csc 510^\circ$ **k** $\sec 510^\circ$ **l** $\cot 510^\circ$

19 Find the exact value of the following trigonometric ratios:

a $\sin \dfrac{2\pi}{3}$ **b** $\cos \dfrac{2\pi}{3}$ **c** $\tan \dfrac{2\pi}{3}$ **d** $\csc \dfrac{2\pi}{3}$

e $\sec \dfrac{2\pi}{3}$ **f** $\cot \dfrac{2\pi}{3}$

20 For each of the following trigonometric ratios:

i Find the value of the reference angle.

ii Find the exact value of the trigonometric ratio.

a $\cos(-240^\circ)$ **b** $\sin 5400^\circ$ **c** $\cos\left(-\dfrac{4\pi}{3}\right)$

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21 Find the exact values of the following:

a $\cos 0^\circ$ **b** $\sin 90^\circ$ **c** $\cos(-180^\circ)$ **d** $\sin 630^\circ$

e $\tan 90^\circ$ **f** $\tan(-630^\circ)$ **g** $\cot 30^\circ$ **h** $\csc 60^\circ$

i $\sec 30^\circ$ **j** $\sec 45^\circ$ **k** $\csc 45^\circ$ **l** $\cot 45^\circ$

m $\cot 270^\circ$ **n** $\sec 180^\circ$ **o** $\csc 0^\circ$ **p** $\cot 450^\circ$

q $\csc 900^\circ$ **r** $\sec(-450)^\circ$

22 Find the exact values of the following:

a $\sin\dfrac{\pi}{2}$ **b** $\sin\left(-\dfrac{\pi}{6}\right)$ **c** $\cos\left(-\dfrac{5\pi}{3}\right)$ **d** $\cos 4\pi$

e $\sin\dfrac{5\pi}{6}$ **f** $\cos\dfrac{3\pi}{4}$ **g** $\sin\dfrac{7\pi}{6}$ **h** $\cos\dfrac{7\pi}{6}$

i $\tan\dfrac{7\pi}{6}$ **j** $\sin\dfrac{5\pi}{3}$ **k** $\cos\dfrac{5\pi}{3}$ **l** $\tan\dfrac{3\pi}{4}$

m $\tan\dfrac{11\pi}{6}$ **n** $\tan\dfrac{3\pi}{2}$ **o** $\sec\dfrac{\pi}{2}$ **p** $\cot 3\pi$

23 By rewriting each ratio in terms of the related acute angle, determine the exact value of the following:

a $$\frac{\sin 120^\circ \cos 240^\circ \tan 330^\circ}{\tan(-45)^\circ}$$

b $$\frac{\left(\sin\dfrac{2\pi}{3}\right)\left(\cos\dfrac{2\pi}{3}\right)\left(\tan\dfrac{3\pi}{4}\right)}{\tan\left(-\dfrac{\pi}{4}\right)}$$

c $$\frac{\sin\left(\dfrac{2\pi}{3}\right)+\cos\left(\dfrac{5\pi}{6}\right)-\tan\left(\dfrac{7\pi}{4}\right)}{\cos\left(\dfrac{4\pi}{3}\right)}$$

d $$\frac{-\sin\left(\dfrac{5\pi}{3}\right)+\cos\left(\dfrac{7\pi}{6}\right)+\tan\left(\dfrac{5\pi}{3}\right)}{-\cos\left(\dfrac{2\pi}{3}\right)}$$

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